



Quiz 4 Online

Date: 01-05-2026

Artificial Neural Networks (ANN)

Student Name: _____ Reg. No: _____

Instructor: Dr Abid Ali

Class: BS AI Red Semester 6th

Total Time: 30 Minutes

Manual Reasoning + Conceptual Understanding

Instructions

- Time allowed: **30 minutes**
- No external tools (no IDE, no ChatGPT, no Google)
- Show **all steps clearly**
- Answers without reasoning will receive **low marks**

Problem Scenario

You are given a **single hidden-layer neural network** used for binary classification.

Network Architecture:

- Input layer: 2 neurons $\rightarrow x_1, x_2$
- Hidden layer: 2 neurons $\rightarrow h_1, h_2$
- Output layer: 1 neuron $\rightarrow y$

Activation Function:

- Hidden layer: **ReLU**
- Output layer: **Sigmoid**

Given Parameters

Input:

$$x_1 = 1.5, x_2 = -2.0$$

Weights & Biases:

Hidden Layer:

$$\begin{aligned} w_{11} &= 0.8, w_{12} = -0.5, b_1 = 0.2 \\ w_{21} &= -1.0, w_{22} = 1.5, b_2 = -0.3 \end{aligned}$$

Output Layer:

$$v_1 = 1.2, v_2 = -0.7, b_o = 0.1$$

◆ Tasks

Part A: Forward Propagation (10 marks)

1. Compute **pre-activation values** for hidden neurons:

$$z_1, z_2$$

2. Apply **ReLU activation**:

$$h_1 = \max(0, z_1), h_2 = \max(0, z_2)$$

3. Compute output neuron input:

$$z_o = v_1 h_1 + v_2 h_2 + b_o$$

4. Apply **Sigmoid function**:

$$y = \frac{1}{1 + e^{-z_o}}$$

Important: Show each step numerically.

Part B: Interpretation (5 marks)

- If threshold = 0.5:
 - What is the predicted class?
 - Explain in **2–3 lines**:
 - Why did the network produce this output?
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Part C: Sensitivity Analysis (5 marks)

Without recomputing everything:

- If x_2 changes from **-2.0 to -1.0**, predict:
 - Will output **increase, decrease, or stay similar**?
 - Justify using **weight signs and ReLU behavior**
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Part D: Critical Thinking (5 marks)

Answer briefly:

1. What happens if ReLU is replaced with **Sigmoid in hidden layer**?
 2. Would the network become:
 - More stable?
 - More prone to vanishing gradients?
 - Explain why
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Part E: Error Analysis (5 marks)

Assume true label = **1**

1. Write the **binary cross-entropy loss formula**
2. Without full calculation:
 - Would loss be **high or low**?
 - Justify using your predicted output